

Product Requirements Document

Automated Camera Trap Triage and Individual Tiger Movement Intelligence System

Pench Tiger Reserve

Document Version: 1.0

Date: August 15, 2026

Classification: Confidential

Prepared For: Pench Tiger Reserve, Forest Department

Domain: Wildlife Conservation & AI/ML

This document contains proprietary information. Unauthorized distribution is prohibited.

Table of Contents

- 1. Executive Summary
- 2. Problem Statement & Background
- 3. Project Objectives
- 4. Stakeholders & Users
- 5. System Overview & Architecture
- 6. Functional Requirements
 - 6.1 Module 1: Blank Image Filtering & Removal
 - 6.2 Module 2: Individual Tiger Identification
 - 6.3 Module 3: Tiger-wise Area Occupancy Mapping
 - 6.4 Module 4: Deviation & Trend Alerting
- 7. Non-Functional Requirements
- 8. Data Architecture & Flow
- 9. Technology Stack
- 10. Literature Review & Research Foundation
- 11. Existing Systems & Competitive Landscape
- 12. Legal, Ethical & Compliance Considerations
- 13. Risk Analysis & Mitigation
- 14. Success Metrics & KPIs
- 15. Project Timeline & Milestones
- 16. Assumptions & Dependencies
- 17. Future Scope & Extensibility
- 18. Glossary of Terms
- 19. References

1. Executive Summary

Pench Tiger Reserve (PTR), spanning over 1,920 sq km across Madhya Pradesh and Maharashtra, is one of India's premier tiger habitats supporting an estimated 40-80+ individual tigers. The reserve deploys hundreds of camera traps across its core (~668 sq km) and buffer (~1,252 sq km) zones, generating millions of images annually. Current manual processing workflows are unsustainable: 70-95% of captured images are blanks triggered by vegetation, wind, or thermal anomalies, and individual tiger identification requires trained experts to laboriously compare stripe patterns across thousands of images.

This PRD defines the requirements for an Automated Camera Trap Triage and Individual Tiger Movement Intelligence System - an end-to-end AI/ML pipeline that will:

- > Automatically filter and quarantine blank images, recovering 70-95% of storage and analyst time.
- > Identify individual tigers through unique stripe-pattern biometrics using deep learning.
- > Build and maintain a persistent, queryable identification database linking individuals to spatiotemporal capture data.
- > Generate per-tiger area occupancy maps with home-range estimates, activity centroids, and territorial overlap visualization.
- > Detect behavioural deviations and raise evidence-backed alerts for range shifts, station intrusions, buffer-zone approaches, and prolonged absences.

The system is designed to transform raw camera-trap data into actionable intelligence for forest department staff, supporting proactive conservation management, anti-poaching operations, and human-wildlife conflict mitigation at Pench Tiger Reserve.

2. Problem Statement & Background

2.1 Core Problem

Camera trapping is the gold standard for non-invasive tiger monitoring, but the sheer volume of data generated overwhelms manual processing capacity. At Pench Tiger Reserve, the key challenges are:

- >> **Data Overload:** Each camera trap station can capture 500-2,000 images per deployment cycle. With 200+ stations, a single survey generates 100,000-400,000 images, of which 70-95% are blank.
- >> **Manual Identification Bottleneck:** Individual tiger identification by stripe-pattern matching is currently performed by a small team of experts, taking weeks to months per survey cycle.
- >> **Delayed Insights:** By the time movement data is analyzed, the intelligence is outdated - range shifts, territorial conflicts, or emergencies may have already escalated.
- >> **No Automated Alerting:** There is no system to proactively flag anomalies such as tigers moving toward village buffer zones, disappearing from regular patrol areas, or exhibiting stress-indicative movement patterns.

2.2 Pench Tiger Reserve Context

- >> **Location:** Straddling Seoni and Chhindwara districts in Madhya Pradesh and Nagpur district in Maharashtra, along the southern slopes of the Satpura Range.
- >> **Total Protected Area:** Approximately 1,920 sq km (MP side: core ~411 sq km, buffer ~768 sq km; Maharashtra side: core ~257 sq km, buffer ~484 sq km).
- >> **Tiger Population:** Estimated 40-80+ individual tigers based on the 2022 All-India Tiger Estimation (Phase IV camera trapping and SECR modeling).
- >> **Monitoring Infrastructure:** Systematic camera trapping grids deployed across Turia, Karmajhiri, Sillari, and other management zones. Data feeds into NTCA's national monitoring framework.
- >> **Key Challenges:** Remote terrain with limited connectivity, seasonal accessibility constraints during monsoons, overlap with human settlements in buffer zones, and rising human-wildlife conflict incidents.

2.3 Current Workflow Limitations

- > Images are manually downloaded from SD cards at each station during periodic field visits.
- > Blank images are visually screened and deleted by field staff - a tedious, error-prone process.
- > Tiger identification relies on side-by-side stripe comparison using printed photographs or basic image viewers.
- > Occupancy and movement analysis is performed retrospectively using spreadsheets and manual GIS plotting.
- > No real-time or near-real-time alerting mechanism exists for anomalous tiger behaviour.
- > Data from MP and Maharashtra sides is often siloed, preventing holistic landscape-level analysis.

3. Project Objectives

3.1 Primary Objectives

- > Automate the classification and quarantine of blank/empty camera trap images with >95% accuracy, reducing manual screening effort by at least 80%.
- > Develop a deep-learning-based individual tiger identification system using stripe-pattern biometrics, capable of matching against a persistent and growing catalogue of known individuals.
- > Build a persistent, queryable database that links each identified individual tiger to every image, camera station, GPS location, and timestamp where it was captured.
- > Generate per-tiger area occupancy maps including capture locations, home-range estimates (using KDE/MCP), activity centroids, estimated occupied area, and inter-individual territorial overlap.
- > Implement an automated deviation and trend alerting engine that compares each processing run against historical baselines and raises evidence-backed alerts for anomalous tiger behaviour.

3.2 Secondary Objectives

- > Reduce the end-to-end survey processing turnaround from weeks/months to hours/days.
- > Provide forest department staff with intuitive, map-based dashboards requiring minimal technical training.
- > Enable landscape-level analysis by integrating data from both the Madhya Pradesh and Maharashtra sides of Pench.
- > Ensure compliance with NTCA data security protocols for sensitive tiger location data.
- > Design the system for extensibility to other species (leopards, wild dogs, sloth bears) and other tiger reserves.
- > Support both cloud-based and on-premise deployment models to accommodate varying connectivity and security requirements.

4. Stakeholders & Users

4.1 Primary Stakeholders

- >> **Field Director, Pench Tiger Reserve:** End decision-maker for reserve management. Requires high-level dashboards, alert summaries, and periodic reports.
- >> **Deputy Directors & Range Officers:** Operational managers who need station-level data, individual tiger tracking, and conflict zone alerts.
- >> **Wildlife Biologists & Researchers:** Require access to raw data, statistical outputs (SECR, occupancy models), and export capabilities for academic analysis.
- >> **Forest Guards & Field Staff:** Upload raw camera trap data and respond to alerts. Need simple, mobile-friendly interfaces.

4.2 Secondary Stakeholders

- >> **National Tiger Conservation Authority (NTCA):** Regulatory body requiring standardized reporting, compliance with M-STrIPES protocols, and access to verified population data.
- >> **Wildlife Institute of India (WII):** Technical partner for monitoring methodology, statistical modeling, and national tiger census coordination.
- >> **State Forest Department (MP & Maharashtra):** Administrative oversight and funding authority.
- >> **Conservation NGOs:** Partners in monitoring, anti-poaching, and community engagement.

5. System Overview & Architecture

5.1 High-Level Architecture

The system follows a modular, pipeline-based architecture with four core processing modules and supporting infrastructure layers. The design adopts a hybrid Edge-to-Cloud model optimized for the remote, low-connectivity environment of a tiger reserve.

5.2 Architectural Layers

- >> **Data Ingestion Layer:** Handles bulk upload of raw image directories from SD cards, network-connected cameras, or edge devices. Supports batch and incremental ingestion with deduplication.
- >> **AI/ML Processing Pipeline:** Sequential processing through: (1) Blank Image Filtering, (2) Tiger Detection & Cropping, (3) Stripe Pattern Extraction, (4) Individual Matching/Enrollment.
- >> **Persistent Data Store:** Relational database (PostgreSQL with PostGIS extension) storing individual profiles, image metadata, GPS coordinates, match confidence scores, and human-review audit trails.
- >> **Spatial Analytics Engine:** Computes per-tiger home ranges (KDE 95%/50% isopleths), activity centroids, area estimates, and territorial overlap matrices. Regenerated on every processing run.
- >> **Deviation Detection & Alerting:** Statistical engine comparing current-run outputs against historical baselines. Generates classified alerts with evidence and confidence levels.
- >> **Visualization & Reporting Layer:** Web-based dashboard with interactive maps, individual tiger profiles, alert feeds, and exportable reports (PDF, CSV, Shapefile).

5.3 Deployment Model

- > Primary deployment: On-premise server at the reserve headquarters with GPU capability (NVIDIA RTX-class or better) for AI inference.
- > Optional cloud tier: AWS/GCP for heavy batch processing, multi-reserve aggregation, and disaster recovery.
- > Edge processing (future): NVIDIA Jetson or Google Coral devices at high-priority stations for real-time blank filtering and alert generation.
- > All sensitive tiger location data remains within government-controlled infrastructure per NTCA protocols.

6. Functional Requirements

6.1 Module 1: Blank Image Filtering & Removal

Purpose: Automatically classify each camera trap frame as blank or containing a subject, quarantine blanks, and report savings.

- >> **FR-1.1 Batch Ingestion:** The system must accept raw image directories (JPEG, PNG, TIFF, BMP) and recursively scan all subdirectories. It must handle directory structures organized by station, date, or camera unit.
- >> **FR-1.2 AI-Based Classification:** Each image must be processed through an object detection model (MegaDetector V6 or equivalent YOLO/RT-DETR architecture) to detect animals, people, and vehicles. Images with no detections above the configurable confidence threshold (default: 0.2) are classified as blank.
- >> **FR-1.3 Confidence Thresholding:** The confidence threshold for blank classification must be configurable by administrators (range: 0.05 - 0.50). Images near the threshold boundary (e.g., within 0.05 of threshold) must be flagged for optional human review.
- >> **FR-1.4 Safe Quarantine:** Blank images must NOT be permanently deleted. They must be moved to a quarantine directory preserving the original folder structure. Quarantine must be reversible - any image can be restored to the working dataset with one action.
- >> **FR-1.5 Processing Report:** After each filtering run, the system must generate a summary report containing: total images scanned, number classified as blank, number retained, confidence distribution histogram, estimated storage saved (MB/GB), estimated processing time saved, and per-station breakdown.
- >> **FR-1.6 Burst/Sequence Handling:** For camera traps operating in burst mode (3-5 rapid frames per trigger), the system must analyse the entire burst and retain the full sequence if any single frame contains a subject.
- >> **FR-1.7 Performance Target:** The blank classifier must achieve a minimum recall of 95% (i.e., miss no more than 5% of true subjects) and a minimum precision of 90% (i.e., no more than 10% of retained images are actually blank).

6.2 Module 2: Individual Tiger Identification via Stripe Pattern

Purpose: Detect tigers in retained images, extract unique stripe-pattern biometrics, match against a persistent catalogue, and build an individual-linked database.

- >> **FR-2.1 Tiger Detection & Cropping:** From each retained image, the system must detect tiger instances using an object detection model (YOLO-based or Faster R-CNN). Each detected tiger must be cropped with appropriate padding (10-20% beyond bounding box) for downstream analysis.
- >> **FR-2.2 Flank Isolation:** The system must identify the flank region (left or right side) of each detected tiger. Flank orientation (L/R) must be recorded as metadata, as stripe patterns differ between sides.
- >> **FR-2.3 Stripe Pattern Extraction:** The system must extract discriminative stripe-pattern features from the isolated flank using deep feature extraction (CNN backbone such as InceptionResNetV2, DenseNet, or a fine-tuned ResNet-50). Features must be stored as embedding vectors in the database.
- >> **FR-2.4 Catalogue Matching:** Each extracted stripe embedding must be compared against all existing individuals in the catalogue using a similarity metric (cosine similarity or learned distance via Siamese/Triplet network). The system must return a ranked list of candidate matches with confidence scores.
- >> **FR-2.5 Automatic vs. Ambiguous Matching:** Matches exceeding a high-confidence threshold (e.g., cosine similarity > 0.85) must be applied automatically. Matches falling in the ambiguous zone (e.g., 0.60 - 0.85) must be surfaced to a human reviewer via a review queue with side-by-side comparison UI. Matches below the low threshold (< 0.60) must trigger new individual enrollment.
- >> **FR-2.6 New Individual Enrollment:** When no confident match is found, the system must create a new individual record with an auto-generated unique identifier (e.g., PTR-T-001), store the reference embedding, and associate all capture metadata.
- >> **FR-2.7 Persistent Database:** The system must maintain a persistent, queryable database (PostgreSQL with PostGIS) linking each identified individual to: all associated images (with file paths), camera station ID, GPS coordinates (latitude/longitude), capture timestamp, flank orientation, match confidence score, and review status (auto-matched / human-verified / pending-review).
- >> **FR-2.8 Catalogue Growth & Re-identification:** The catalogue must grow incrementally across processing runs. Previously identified individuals must be re-identified in new captures. The system must handle at least 200 individual tiger profiles with 10,000+ associated images without performance degradation.
- >> **FR-2.9 Human Review Interface:** Ambiguous matches must be presented in a web-based review interface showing: the query image, top-K candidate matches from the catalogue, side-by-side flank comparison, confidence scores, and accept/reject/merge actions for the reviewer.

6.3 Module 3: Tiger-wise Area Occupancy Mapping

Purpose: After each processing run, generate per-tiger spatial occupancy analysis including capture locations, home ranges, centroids, and territorial overlap - visualized on a reserve map.

- >> **FR-3.1 Per-Tiger Capture Location Set:** For every identified individual, the system must compile the complete set of capture locations (GPS coordinates) with associated timestamps, forming a spatiotemporal capture history.
- >> **FR-3.2 Home Range Estimation:** The system must compute home-range estimates for each individual using Kernel Density Estimation (KDE) at 95% (total range) and 50% (core area) utilization distribution isopleths. Minimum Convex Polygon (MCP) estimates must also be provided as an alternative method.
- >> **FR-3.3 Activity Centroid:** For each individual, compute the weighted centroid of activity based on capture frequency and recency. The centroid must be plotted on the reserve map.
- >> **FR-3.4 Area Estimation:** The system must calculate the estimated area occupied by each individual (in sq km) from the KDE 95% isopleth. Summary statistics (mean, median, min, max across all individuals) must be reported.
- >> **FR-3.5 Territorial Overlap Analysis:** The system must compute pairwise overlap indices between individuals' home ranges using volume-of-intersection or Bhattacharyya's affinity. Overlap must be visualized as a heatmap matrix and as overlaid polygons on the map.
- >> **FR-3.6 Map Visualization:** All spatial outputs must be rendered on an interactive map of the Pench Tiger Reserve showing: reserve boundary (core and buffer), camera station locations, per-tiger KDE polygons (color-coded by individual), activity centroids, and overlap zones. The map must support zoom, pan, layer toggling, and individual tiger selection.
- >> **FR-3.7 Export Formats:** Occupancy outputs must be exportable as: GeoJSON/Shapefile (for GIS software), PDF maps (for printed reports), CSV data tables (for spreadsheet analysis), and KML (for Google Earth visualization). These formats are critical for forest department staff who may use diverse tools.
- >> **FR-3.8 Run-by-Run Regeneration:** All occupancy outputs must be regenerated from scratch on every processing run, reflecting the latest cumulative data. Historical snapshots of previous runs must be preserved for comparison.

6.4 Module 4: Deviation & Trend Alerting

Purpose: Compare each processing run against established individual histories and raise classified, evidence-backed alerts for meaningful behavioural deviations.

- >> **FR-4.1 Historical Baseline:** The system must maintain a per-individual historical profile including: established home range, regularly visited stations, visit frequency patterns, seasonal movement patterns, and last-seen timestamp.
- >> **FR-4.2 Range Shift Detection:** The system must detect and alert when an individual's current home-range centroid has shifted beyond defined thresholds: >15-20 sq km in the core zone or >5 km linear displacement in the buffer zone. The alert must include the magnitude and direction of the shift.
- >> **FR-4.3 Novel Station Alert:** An alert must be raised when an individual is captured for the first time at a camera station not previously within its established range. The alert must indicate the distance from the individual's known range boundary.
- >> **FR-4.4 Buffer/Village Proximity Alert:** The system must flag when an individual's captures indicate movement into or toward buffer-zone or village-adjacent stations. This alert is critical for human-wildlife conflict early warning.
- >> **FR-4.5 Prolonged Absence Alert:** When a previously regular individual (defined as captured in ≥ 3 of the last 5 survey cycles) is not detected in the current run, a prolonged-absence alert must be raised. The alert must include: last known location, days since last capture, and historical capture frequency.
- >> **FR-4.6 Artefact Filtering:** The alerting engine must distinguish genuine behavioural deviations from survey artefacts, including: camera station malfunction (no images from any species), seasonal trail inaccessibility, incomplete survey coverage, and known camera relocation. Artefact-related non-detections must not generate false alerts.
- >> **FR-4.7 Alert Classification:** Each alert must include: alert type (range shift / novel station / buffer approach / prolonged absence), affected individual ID and name (if assigned), detected change (quantified), supporting evidence (images, locations, timestamps), confidence level (High / Medium / Low), and recommended action.
- >> **FR-4.8 Alert Delivery:** Alerts must be delivered via: in-dashboard notification feed, email digest to configured recipients, SMS for high-priority alerts (buffer/village proximity), and exportable alert reports (PDF).
- >> **FR-4.9 Trend Analysis:** Beyond single-run deviations, the system must support multi-run trend analysis showing: population trajectory (new individuals vs. lost individuals over time), territory fragmentation or consolidation trends, seasonal migration patterns, and camera network coverage effectiveness.

7. Non-Functional Requirements

7.1 Performance

- > Blank image filtering must process at least 1,000 images per minute on recommended hardware (GPU-equipped server).
- > Individual tiger identification must process a single image (detection through matching) in under 5 seconds.
- > Occupancy map regeneration for up to 80 individuals must complete within 30 minutes.
- > The system must handle datasets of up to 500,000 images per survey cycle without degradation.

7.2 Scalability

- > The individual catalogue must support at least 200 tigers with 50,000+ associated images.
- > The system must be extensible to process data from multiple tiger reserves without architectural changes.
- > Database storage must scale linearly with ingested data volume.

7.3 Reliability & Availability

- > The system must operate reliably with intermittent internet connectivity (core processing must be offline-capable).
- > All processing operations must be idempotent - re-running a batch must produce consistent results.
- > Quarantined images and database records must never be permanently deleted without explicit administrator action and dual confirmation.

7.4 Security

- > All tiger GPS location data must be encrypted at rest (AES-256) and in transit (TLS 1.3).
- > Access must be role-based (RBAC): Administrator, Biologist, Range Officer, Field Staff, and Read-Only Viewer.
- > Audit logs must record all data access, modifications, exports, and alert acknowledgments.
- > The system must comply with NTCA data security protocols for the National Repository of Camera Trap Photographs of Tigers (NRCTPT).

7.5 Usability

- > The dashboard must be accessible via modern web browsers (Chrome, Firefox, Edge) without plugin installation.
- > Key workflows (data upload, alert review, map exploration) must be completable with minimal training (<4 hours).
- > The interface must support both English and Hindi labels for field-staff accessibility.
- > Map visualizations must be usable on tablets for field use.

7.6 Maintainability

- > AI models must be re-trainable with new data without system downtime (model hot-swapping).

- > Configuration (thresholds, alert rules, camera station metadata) must be adjustable via an admin panel without code changes.
- > Comprehensive API documentation must be maintained for all system interfaces.

8. Data Architecture & Flow

8.1 Data Flow Pipeline

The system processes camera trap data through a sequential pipeline with five stages:

- >> **Stage 1 - Ingestion:** Raw image directories are uploaded via batch upload interface or file-system mount. EXIF metadata (timestamp, camera model) and station metadata (GPS, station ID) are extracted and indexed.
- >> **Stage 2 - Blank Filtering:** MegaDetector V6 processes each image. Blank images (no animal/person/vehicle detected above threshold) are moved to quarantine. Retained images proceed to Stage 3.
- >> **Stage 3 - Tiger Detection:** Retained images are processed by a tiger-specific detector (fine-tuned YOLO). Detected tigers are cropped with flank isolation. Non-tiger wildlife images are tagged and catalogued but not processed further in this pipeline.
- >> **Stage 4 - Identity Matching:** Flank embeddings are extracted via CNN backbone and compared against the catalogue using Siamese network similarity scoring. Automatic matches, ambiguous matches (human review queue), and new enrollments are processed accordingly.
- >> **Stage 5 - Spatial Analytics & Alerting:** Updated individual records feed into the KDE home-range engine and deviation detection module. Maps, occupancy stats, and alerts are generated and published to the dashboard.

8.2 Database Schema (Key Entities)

- >> **Individuals Table:** Individual ID (PK), assigned name, first capture date, total captures count, last seen timestamp, primary flank embedding vector, status (active/absent/unconfirmed).
- >> **Captures Table:** Capture ID (PK), individual ID (FK), image file path, station ID (FK), GPS latitude/longitude, capture timestamp, flank side (L/R), match confidence, review status, reviewer ID, review timestamp.
- >> **Stations Table:** Station ID (PK), station name, GPS coordinates, zone (core/buffer), nearest village, deployment status, last maintenance date.
- >> **Alerts Table:** Alert ID (PK), alert type, individual ID (FK), detected change description, evidence (JSON), confidence level, status (new/acknowledged/resolved), assigned officer, creation timestamp.
- >> **Processing Runs Table:** Run ID (PK), start/end timestamps, total images processed, blanks removed, tigers identified, new individuals enrolled, alerts generated, operator ID.

8.3 Data Standards & Interoperability

- > Image metadata must follow the Camtrap DP (Camera Trap Data Package) standard for interoperability with global platforms like Wildlife Insights and LILA BC.
- > Spatial data must use WGS 84 (EPSG:4326) coordinate reference system.
- > All timestamps must be stored in UTC with IST conversion for display.
- > Export formats must include COCO Camera Traps annotation format for ML community compatibility.

9. Technology Stack

9.1 AI/ML Frameworks

- > Object Detection: MegaDetector V6 (YOLOv9/YOLOv10/RT-DETR variants) via PyTorch-Wildlife package for blank filtering and animal detection.
- > Tiger-Specific Detection: Fine-tuned YOLOv8/v9 trained on Indian tiger camera trap datasets.
- > Individual Identification: Siamese Network with Triplet Loss using InceptionResNetV2 or DenseNet backbone for stripe-pattern embedding extraction.
- > Spatial Analytics: Python scipy, scikit-learn for KDE; R adehabitatHR and secr packages for advanced SECR modeling.

9.2 Backend Infrastructure

- > Runtime: Python 3.10+ with FastAPI for REST API layer.
- > Database: PostgreSQL 15+ with PostGIS extension for spatial queries.
- > Vector Store: pgvector extension for efficient embedding similarity search.
- > Task Queue: Celery with Redis for asynchronous image processing pipeline management.
- > Object Storage: MinIO (on-premise S3-compatible) for image storage with lifecycle policies.

9.3 Frontend & Visualization

- > Dashboard: React.js or Next.js web application with responsive design.
- > Mapping: Leaflet.js or Mapbox GL JS for interactive reserve maps with GeoJSON overlay.
- > Charts: D3.js or Recharts for population trends, overlap heatmaps, and alert timelines.
- > Reporting: Server-side PDF generation (ReportLab/WeasyPrint) for forest department reports.

9.4 Hardware Requirements

- > Processing Server: 32+ GB RAM, NVIDIA GPU (RTX 3090/4090 or A4000 with 16+ GB VRAM), 2+ TB NVMe SSD storage.
- > Database Server: 16+ GB RAM, 1+ TB storage (can be co-located with processing server for smaller deployments).
- > Edge Devices (optional): NVIDIA Jetson Orin Nano or Google Coral Dev Board for station-level pre-processing.
- > Networking: Local gigabit Ethernet; cellular/satellite uplink for remote alert transmission.

10. Literature Review & Research Foundation

The system design is grounded in extensive peer-reviewed research across computer vision, wildlife ecology, and conservation technology. Key research areas and findings are summarized below:

10.1 Blank Image Filtering & Camera Trap AI

- > MegaDetector (Microsoft AI for Earth, Beery et al., 2019): The industry-standard open-source model for camera trap image triage. Uses YOLO-family architectures to detect animals, people, and vehicles. Achieves >82% animal recall across diverse ecosystems. MegaDetector V6 (2024) introduced compact variants with YOLOv9 and RT-DETR backbones for edge deployment.
- > Deep learning surpasses traditional approaches: Transfer learning with ResNet-50/101 and EfficientNet architectures achieves >95% blank classification accuracy, significantly outperforming pixel-difference and HOG+SVM methods (Norouzzadeh et al., 2018; Tabak et al., 2019).
- > Cascade architecture: Running an object detector (MegaDetector) before a species classifier dramatically reduces computational load and false-positive rate - a two-stage pipeline is now the accepted best practice in camera trap AI.

10.2 Individual Animal Re-Identification

- > Tiger stripe uniqueness: Tiger stripe patterns are individually unique - analogous to human fingerprints - and remain stable throughout an individual's life (Karanth & Nichols, 1998). This biological property enables photographic capture-recapture studies.
- > HotSpotter (Crall et al., 2013): A SIFT-based keypoint matching algorithm originally developed for zebra and giraffe identification. Extracts local distinctive features ('hotspots') from body patterns. Effective but struggles with significant pose variation and partial occlusion.
- > Deep metric learning: Siamese Networks with Triplet Loss learn an embedding space where same-individual images are close and different-individual images are distant. CNN backbones (InceptionResNetV2, DenseNet-169) extract robust features invariant to pose, lighting, and partial occlusion (Li et al., 2019; Schneider et al., 2020).
- > TigerNet and domain-specific architectures: Specialized networks use adaptive feature enhancement to isolate stripe topography from background noise and employ Cosine Loss for verification. These achieve 92-97% rank-1 accuracy on curated datasets (Sharma et al., 2022).

10.3 Spatial Ecology & Home Range Analysis

- > Kernel Density Estimation (KDE): The standard method for utilization distribution estimation. 95% isopleths define total home range; 50% isopleths define core areas. Autocorrelated KDE (AKDE) accounts for temporal dependence in relocation data (Fleming et al., 2015).
- > Spatially Explicit Capture-Recapture (SECR): Combines individual identification from camera traps with spatial capture locations to jointly estimate population density and individual home-range centers (Efford, 2004; Royle et al., 2014). Implemented in R packages 'secr' and 'oSCR'.
- > Camera trap occupancy modeling: Detection/non-detection data across spatial grids estimates species presence probability while accounting for imperfect detection (MacKenzie et al., 2002). Essential for

landscape-level distribution mapping.

- > Tiger territory sizes in Central Indian reserves: Male tigers typically occupy 40-100 sq km; females 15-40 sq km. Territories overlap more at reserve edges and buffer zones (Jhala et al., 2020; Karanth et al., 2006).

10.4 Anomaly Detection & Movement Analysis

- > Change-point detection: Statistical methods (PELT, Bayesian Online Change Point Detection) identify abrupt shifts in time-series movement data, enabling automated detection of range shifts or behavioural state changes (Killick et al., 2012).
- > LSTM-based trajectory analysis: Long Short-Term Memory networks model sequential movement patterns and flag anomalous trajectories that deviate from learned normal behaviour (Wijayakulasooriya et al., 2020).
- > Edge AI for real-time alerting: Lightweight models (YOLO-family) on NVIDIA Jetson devices enable sub-second detection and alert generation at camera stations, overcoming remote connectivity constraints (TrailGuard AI, RESOLVE).

11. Existing Systems & Competitive Landscape

The proposed system builds upon and differentiates from several existing platforms in the conservation technology space:

- >> **Wildlife Insights (Google):** Cloud-based platform for camera trap data management. Uses SpeciesNet (TensorFlow) for species-level classification at scale (3.6M images/hour). Strengths: massive scalability, global biodiversity dashboards. Limitations: no individual-level identification, no tiger stripe matching, no movement alerting. SpeciesNet was open-sourced in 2025.
- >> **MegaDetector (Microsoft AI for Earth):** Open-source object detection model for camera trap triage. Strengths: industry-standard blank filtering, high accuracy, active community. Limitations: detection only (no species classification or individual ID). Our system uses MegaDetector as the first stage in the pipeline.
- >> **M-STRIPES (NTCA/WII):** India's official tiger monitoring system. Android app for field patrols and GIS-based desktop software for ecological analysis. Strengths: regulatory standard, nationwide deployment, field-tested. Limitations: no AI-based image processing, no automated individual identification, primarily a patrol-logging and reporting tool.
- >> **TrailGuard AI (RESOLVE/Intel):** Edge AI camera hardware for real-time intruder detection and alerting. Strengths: sub-30-second alert delivery, ruggedized hardware, solar-powered. Limitations: hardware-focused, no individual identification, no spatial analytics.
- >> **SMART Conservation Tools:** Open-source patrol management and analytics platform used in 800+ sites globally. Strengths: comprehensive patrol planning, incident reporting, spatial dashboards. Limitations: no camera trap AI integration, no automated identification.
- >> **HotSpotter / WildID / ExtractCompare:** Desktop tools for pattern-based individual identification. Strengths: proven in research contexts, open-source (HotSpotter). Limitations: manual workflow, no pipeline integration, limited scalability, no spatial analytics or alerting.

Differentiation: The proposed system uniquely integrates all four capabilities - blank filtering, individual identification, spatial occupancy, and deviation alerting - into a single automated pipeline tailored for Indian tiger reserves, with compliance to NTCA protocols and M-STRIPES data standards.

12. Legal, Ethical & Compliance Considerations

12.1 Data Security & NTCA Compliance

- > Tiger GPS location data is classified as sensitive under NTCA guidelines due to poaching risks. Precise coordinates must not be exposed publicly or shared without explicit authorization from the Field Director.
- > All data must be stored in government-controlled infrastructure. Cloud processing, if used, must employ isolated, encrypted environments with no third-party data access.
- > The system must support data contribution to the National Repository of Camera Trap Photographs of Tigers (NRCTPT) in the required format.
- > Access control must align with State Forest Department hierarchies and NTCA-approved data sharing protocols.

12.2 Human Privacy Protection

- > Camera traps frequently capture images of local villagers, forest staff, and visitors. These incidental human captures must be handled with care.
- > The system must implement automatic face-blurring or human-image flagging in the pipeline, segregating human-containing images from the wildlife processing stream.
- > Human images indicating potential illegal activity (poaching, encroachment) must be routed to a secure, access-restricted review queue for authorized personnel only.
- > Local communities in buffer zones should be informed about camera trap presence as part of the reserve's community engagement protocol.

12.3 Ethical AI Considerations

- > Model decisions (blank classification, individual ID matching) must be auditable. Confidence scores and supporting evidence must be logged for every automated decision.
- > The system must not make irreversible decisions (permanent image deletion, individual merger) without human confirmation.
- > AI model biases (e.g., lower accuracy for night images, cubs, or partially occluded tigers) must be documented and communicated to users.
- > Model performance must be continuously monitored and reported, with automatic flagging if accuracy degrades below defined thresholds.

12.4 Regulatory Framework

- > Wildlife Protection Act, 1972 (India): Governs all activities in tiger reserves. System deployment requires necessary permissions from the State Chief Wildlife Warden.
- > National Tiger Conservation Authority Act, 2006: Defines monitoring obligations for tiger reserves, including data reporting requirements.
- > Forest Rights Act, 2006: Relevant for buffer-zone operations and community-related data handling.
- > IT Act, 2000 & Digital Personal Data Protection Act, 2023: Applicable to any personal data (human images, staff credentials) processed by the system.

13. Risk Analysis & Mitigation

>> Risk: Domain Shift in AI Models:

- Description: MegaDetector and species classifiers may exhibit 10-60% accuracy drops when deployed in new geographic regions with unfamiliar vegetation, lighting, and camera angles.
- Mitigation: Fine-tune models on Pench-specific training data (minimum 5,000 labeled images). Implement continuous accuracy monitoring with automatic flagging if performance drops below thresholds.

>> Risk: Insufficient Training Data for Individual ID:

- Description: New or rarely photographed individuals may have insufficient reference images for reliable stripe-pattern matching.
- Mitigation: Implement a confidence-tiered matching system. Low-confidence matches are queued for human review. Minimum 3 reference images per flank required before auto-matching is enabled for an individual.

>> Risk: Connectivity Limitations:

- Description: Remote camera stations have no cellular or internet connectivity, limiting real-time data transmission.
- Mitigation: Design core processing to be fully offline-capable. Edge processing (future phase) can pre-filter at the station. Batch uploads during periodic field visits remain the primary data flow.

>> Risk: Hardware Failure in Remote Conditions:

- Description: Processing servers at reserve headquarters may be affected by power outages, heat, humidity, or monsoon-related damage.
- Mitigation: Deploy UPS and surge protection. Maintain automated backups to secondary storage. Design the system for graceful degradation and recovery.

>> Risk: Staff Adoption & Training:

- Description: Forest department staff may resist or struggle to adopt new technology due to limited digital literacy.
- Mitigation: Design for simplicity - key workflows must require <4 hours of training. Provide Hindi language support. Assign dedicated on-site champions for the first 6 months.

>> Risk: False Alerts Causing Alert Fatigue:

- Description: Excessive or inaccurate deviation alerts may cause staff to ignore genuinely critical notifications.
- Mitigation: Implement artefact filtering to suppress false alerts from camera malfunctions or survey gaps. Provide configurable alert thresholds and daily/weekly digest modes.

>> Risk: Data Security Breach:

- Description: Unauthorized access to sensitive tiger location data could facilitate poaching.
- Mitigation: Encrypt all data at rest and in transit. Implement strict RBAC, audit logging, and network isolation. Conduct periodic security audits. No public-facing endpoints.

>> Risk: Model Obsolescence:

- Description: Rapid advances in AI may render current model architectures outdated within 2-3 years.
- Mitigation: Design modular model-serving architecture allowing hot-swap of detection and identification models without system rebuild. Maintain abstraction layers between pipeline logic and model inference.

14. Success Metrics & KPIs

14.1 Processing Efficiency Metrics

- > Blank image detection accuracy: >95% recall, >90% precision.
- > Processing throughput: >1,000 images/minute for blank filtering; <5 seconds/image for individual identification.
- > End-to-end survey processing time: Reduced from weeks to <48 hours for a full survey cycle of 200,000 images.
- > Storage savings: >70% reduction in working dataset size after blank quarantine.

14.2 Identification Accuracy Metrics

- > Individual tiger re-identification: >90% Rank-1 accuracy (correct individual in top match).
- > Top-5 accuracy: >97% (correct individual within top 5 candidates).
- > False enrollment rate: <5% (new individual created when the tiger already exists in catalogue).
- > Human review queue volume: <15% of total tiger detections require manual review.

14.3 Ecological Output Metrics

- > Home range estimation correlation: >0.85 Pearson correlation with expert-generated home ranges using the same data.
- > Individual coverage: >90% of known individuals in the reserve successfully enrolled and tracked.
- > Alert precision: >80% of generated alerts validated as genuine by domain experts (not survey artefacts).

14.4 Operational Impact Metrics

- > Analyst time savings: >80% reduction in manual image screening and identification effort.
- > Alert response time: <24 hours from data upload to alert delivery for high-priority deviations.
- > User adoption: >80% of authorized staff regularly using the dashboard within 6 months of deployment.
- > Data completeness: >95% of camera trap images from each survey cycle successfully ingested and processed.

15. Project Timeline & Milestones

>> Phase 1: Discovery & Data Collection (Months 1-2):

- Stakeholder interviews and requirement validation with PTR field staff.
- Collection and annotation of Pench-specific training dataset (minimum 10,000 images).
- Camera station metadata compilation (GPS, zone classification, deployment history).
- Infrastructure assessment and hardware procurement.
- Data sharing agreements with NTCA and State Forest Department.

>> Phase 2: Core Pipeline Development (Months 3-5):

- MegaDetector V6 integration and fine-tuning for Pench ecosystem.

- Blank image filtering module with quarantine workflow.
- Tiger detection and flank isolation model training.
- Stripe-pattern embedding extraction pipeline.
- Database schema implementation (PostgreSQL + PostGIS + pgvector).

>> Phase 3: Identification & Matching Engine (Months 5-7):

- Siamese network training for individual re-identification.
- Catalogue matching and new-individual enrollment logic.
- Human review interface for ambiguous matches.
- Integration testing with historical Pench camera trap data.
- Initial catalogue seeding with known individuals.

>> Phase 4: Spatial Analytics & Alerting (Months 7-9):

- KDE home-range estimation engine implementation.
- Interactive map dashboard with reserve boundaries and layers.
- Territorial overlap computation and visualization.
- Deviation detection engine with configurable alert rules.
- Alert delivery system (dashboard, email, SMS).

>> Phase 5: Testing, Training & Deployment (Months 9-11):

- End-to-end system testing with full survey dataset.
- Performance benchmarking and optimization.
- Security audit and NTCA compliance review.
- Staff training workshops (English and Hindi).
- Staged rollout: pilot with 2 management zones, then full reserve.

>> Phase 6: Monitoring & Iteration (Months 11-12+):

- Production monitoring and accuracy tracking.
- Model refinement based on field feedback and new data.
- Feature enhancements based on user requests.
- Documentation finalization and knowledge transfer.
- Handover and long-term maintenance plan establishment.

16. Assumptions & Dependencies

16.1 Assumptions

- > Camera trap images will be provided in standard formats (JPEG, PNG) with intact EXIF metadata including timestamps.
- > Camera station GPS coordinates and zone classifications will be provided by the reserve management.
- > A minimum of 5,000 labeled images from Pench will be available for model fine-tuning, including blank/non-blank labels and species-level annotations.
- > At least 30 known individual tigers with multiple reference images will be available for initial catalogue seeding.
- > The reserve headquarters has reliable power supply (with backup) and physical security for server hardware.
- > Forest department staff are willing to participate in training and adopt the new workflow.
- > Internet connectivity (even if intermittent) is available at headquarters for software updates, cloud sync, and alert delivery.

16.2 Dependencies

- > Availability of GPU-equipped server hardware at the reserve (procurement timeline: 4-6 weeks).
- > NTCA approval for data processing infrastructure and data-sharing protocols.
- > State Forest Department approval for system deployment and staff allocation.
- > Access to historical camera trap datasets from previous survey cycles for training and baseline establishment.
- > Coordination between MP and Maharashtra PTR administrations for cross-border data integration.
- > Continued maintenance of camera trap infrastructure by field staff (battery replacement, SD card swaps, camera repositioning).

17. Future Scope & Extensibility

17.1 Species Extension

- > Leopard identification using rosette patterns (similar Siamese network approach).
- > Wild dog (Dhole) pack identification and tracking.
- > Prey species density estimation (Chital, Sambar, Nilgai) for prey-predator dynamics analysis.
- > Sloth bear, Indian Gaur, and other Schedule-I species monitoring.

17.2 Technical Enhancements

- > Real-time processing via edge AI deployment at high-priority camera stations (NVIDIA Jetson Orin).
- > Video analysis support (motion detection, behaviour classification from video clips).
- > Integration with collar/GPS telemetry data for combined camera-trap + telemetry movement analysis.

- > Satellite imagery integration for habitat change detection and corridor connectivity analysis.
- > Automated report generation aligned with NTCA's Annual Tiger Reserve Report format.

17.3 Multi-Reserve Scaling

- > Deployment to other Central Indian tiger reserves (Kanha, Satpura, Tadoba, Melghat).
- > Cross-reserve individual tracking for dispersing tigers (landscape-level connectivity).
- > Federated learning across reserves to improve model accuracy without centralizing sensitive data.
- > National dashboard for NTCA aggregating insights across all participating reserves.

18. Glossary of Terms

- >> **AKDE:** Autocorrelated Kernel Density Estimation - a home-range estimation method that accounts for temporal autocorrelation in movement data.
- >> **Camtrap DP:** Camera Trap Data Package - a standardized data exchange format for camera trap studies.
- >> **CNN:** Convolutional Neural Network - a class of deep learning models particularly effective for image recognition tasks.
- >> **EXIF:** Exchangeable Image File Format - metadata embedded in image files containing camera settings, timestamps, and GPS data.
- >> **KDE:** Kernel Density Estimation - a non-parametric method for estimating the probability density function, used here for home-range estimation.
- >> **LILA BC:** Labeled Information Library of Alexandria: Biology and Conservation - a public repository of labeled camera trap datasets.
- >> **MCP:** Minimum Convex Polygon - the smallest convex polygon enclosing all capture points, used as a simple home-range estimate.
- >> **MegaDetector:** An open-source AI model by Microsoft AI for Earth that detects animals, people, and vehicles in camera trap images.
- >> **M-STRIPES:** Monitoring System for Tigers: Intensive Protection and Ecological Status - NTCA's official tiger monitoring IT system.
- >> **NTCA:** National Tiger Conservation Authority - the statutory body governing tiger conservation in India.
- >> **PostGIS:** A spatial database extension for PostgreSQL enabling geographic object storage and analysis.
- >> **PTR:** Pench Tiger Reserve.
- >> **RBAC:** Role-Based Access Control - a security mechanism restricting system access based on user roles.
- >> **SECR:** Spatially Explicit Capture-Recapture - a statistical framework estimating population density from individual identification and spatial capture data.
- >> **Siamese Network:** A neural network architecture containing two identical sub-networks used for comparing similarity between two inputs.
- >> **Triplet Loss:** A loss function used in metric learning that ensures an anchor is closer to a positive example than a negative example in embedding space.
- >> **UD:** Utilization Distribution - a probability surface representing the relative frequency of an animal's use of space.
- >> **WII:** Wildlife Institute of India - a research and training institution under the Ministry of Environment, Forest and Climate Change.
- >> **YOLO:** You Only Look Once - a family of real-time object detection neural network architectures.

19. References

- [1] Beery, S., Morris, D., & Yang, S. (2019). Efficient Pipeline for Camera Trap Image Review. arXiv preprint arXiv:1907.06772. Microsoft AI for Earth.
- [2] Crall, J. P., Stewart, C. V., Berger-Wolf, T. Y., Rubenstein, D. I., & Sundaresan, S. R. (2013). HotSpotter - Patterned Species Instance Recognition. IEEE Workshop on Applications of Computer Vision (WACV).
- [3] Efford, M. G. (2004). Density estimation in live-trapping studies. *Oikos*, 106(3), 598-610.
- [4] Fleming, C. H., Fagan, W. F., Mueller, T., Olson, K. A., Leimgruber, P., & Calabrese, J. M. (2015). Rigorous home range estimation with movement data: A new autocorrelated kernel density estimator. *Ecology*, 96(5), 1182-1188.
- [5] Jhala, Y. V., Qureshi, Q., & Nayak, A. K. (2020). Status of Tigers, Co-predators and Prey in India 2018. National Tiger Conservation Authority & Wildlife Institute of India, TR No. 2020/05.
- [6] Jhala, Y. V., Qureshi, Q., & Nayak, A. K. (2022). Status of Tigers, Co-predators and Prey in India: Summary Report 2022. National Tiger Conservation Authority & Wildlife Institute of India.
- [7] Karanth, K. U., & Nichols, J. D. (1998). Estimation of tiger densities in India using photographic captures and recaptures. *Ecology*, 79(8), 2852-2862.
- [8] Karanth, K. U., Nichols, J. D., Kumar, N. S., & Hines, J. E. (2006). Assessing tiger population dynamics using photographic capture-recapture sampling. *Ecology*, 87(11), 2925-2937.
- [9] Killick, R., Fearnhead, P., & Eckley, I. A. (2012). Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500), 1590-1598.
- [10] Li, S., Li, J., Tang, H., Qian, R., & Lin, W. (2019). ATRW: A Benchmark for Amur Tiger Re-identification in the Wild. ACM International Conference on Multimedia.
- [11] MacKenzie, D. I., Nichols, J. D., Lachman, G. B., Droege, S., Royle, J. A., & Langtimm, C. A. (2002). Estimating site occupancy rates when detection probabilities are less than one. *Ecology*, 83(8), 2248-2255.
- [12] Norouzzadeh, M. S., Nguyen, A., Kosmala, M., Swanson, A., Palmer, M. S., Packer, C., & Clune, J. (2018). Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning. *PNAS*, 115(25), E5716-E5725.
- [13] Royle, J. A., Chandler, R. B., Sollmann, R., & Gardner, B. (2014). *Spatial Capture-Recapture*. Academic Press.
- [14] Schneider, S., Taylor, G. W., Linqvist, S., & Kremer, S. C. (2020). Past, present and future approaches using computer vision for animal re-identification from camera trap data. *Methods in Ecology and Evolution*, 11(5), 619-636.
- [15] Tabak, M. A., Norouzzadeh, M. S., Wolfson, D. W., Sweeney, S. J., VerCauteren, K. C., Snow, N. P., ... & Miller, R. S. (2019). Machine learning to classify animal species in camera trap images. *Methods in Ecology and Evolution*, 10(4), 585-590.
- [16] RESOLVE. TrailGuard AI: Artificial Intelligence for Wildlife Protection. <https://www.resolve.ngo/trailguard.htm>
- [17] Google. Wildlife Insights: A platform to manage, analyze, and share camera trap data. <https://www.wildlifeinsights.org>
- [18] Microsoft AI for Earth. MegaDetector: AI for camera trap image processing. <https://github.com/microsoft/CameraTraps>
- [19] SMART Conservation Tools. Spatial Monitoring and Reporting Tool. <https://smartconservationtools.org>

--- End of Document ---

Product Requirements Document v1.0
Automated Camera Trap Triage & Tiger Movement Intelligence System
Pench Tiger Reserve

*This document is confidential and intended solely for authorized personnel.
Unauthorized reproduction or distribution is strictly prohibited.*